Lecture Scribe: Joint Distribution of N Random Variables

CSE400 – Lecture 19

1 Definitions and Notation

1.1 Joint Distribution of N Random Variables

For random variables $X_1, X_2, \ldots, X_N$, the vector form is:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

The joint distribution describes the probability behavior of all variables together. It can be represented using:

- Joint PMF
- Joint PDF
- Joint CDF

1.2 Joint PMF

$$P_{X_1,\ldots,X_N}(x_1, \ldots, x_N) = P(X_1 = x_1, \ldots, X_N = x_N)$$

1.3 Joint CDF

$$F_{X_1,\ldots,X_N}(x_1, \ldots, x_N) = P(X_1 \le x_1, \ldots, X_N \le x_N)$$

1.4 Joint PDF

$$f_{X_1,\ldots,X_N}(x_1,\ldots,x_N) = \frac{\partial^N F_{X_1,\ldots,X_N}}{\partial x_1 \cdots \partial x_N}$$

2 Jointly Gaussian Random Variables

2.1 Joint Gaussian PDF

$$\begin{aligned} f_{XY}(x,y) = & \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho_{XY}^2}} \\ & \times \exp\left(-\frac{1}{2(1-\rho_{XY}^2)}\left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2\right.\right. \\ & \left.\left. -2\rho_{XY}\left(\frac{x-\mu_x}{\sigma_x}\right)\left(\frac{y-\mu_y}{\sigma_y}\right) + \left(\frac{y-\mu_y}{\sigma_y}\right)^2\right]\right) \end{aligned}$$

2.2 Correlation Coefficient

$$\rho_{XY} = \frac{\mathrm{cov}(X,Y)}{\sigma_X\sigma_Y}$$

3 Marginal Distributions

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx$$

3.1 Key Property

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

4 Independence of Random Variables

$$f_{X_1,\ldots,X_N}(x_1,\ldots,x_N) = \prod_{i=1}^{N} f_{X_i}(x_i)$$

5 Worked Examples

5.1 Example 1: Joint Gaussian

Given:

$$\mu_X = 2, \quad \sigma_X = 1, \quad \mu_Y = -3, \quad \sigma_Y = 2, \quad \text{cov}(X, Y) = -1$$

5.1.1 (a) Correlation Coefficient

$$\rho_{XY} = \frac{-1}{(1)(2)} = -\frac{1}{2}$$

5.1.2 (b) Joint PDF

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sqrt{3}} \exp\left(-\frac{2}{3}\left[(x-2)^2 + \frac{(x-2)(y+3)}{2} + \frac{(y+3)^2}{4}\right]\right)$$

5.1.3 (c) Marginal PDFs

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-2)^2}{2}}$$

$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} e^{-\frac{(y+3)^2}{8}}$$

5.1.4 (d) Probability

$$Z = \frac{X-2}{1}$$

$$P(X < 0) = P(Z < -2) = P(Z > 2) = Q(2)$$

5.2 Example 2: Uniform 3D Vector

$$f_{X_1,X_2,X_3} = \begin{cases} C, & |x_i| \le 1 \\ 0, & \text{otherwise} \end{cases}$$

5.2.1 (a) Constant

$$C(2)(2)(2) = 1 \Rightarrow C = \frac{1}{8}$$

5.2.2 (b) Marginal

$$f_{X_1,X_2} = \frac{1}{4}$$

5.2.3 (c) Marginal

$$f_{X_1} = \frac{1}{2}$$

5.2.4 (d) Conditional

$$f_{X_1|X_2,X_3} = \frac{1/8}{1/4} = \frac{1}{2}$$

5.2.5 (e) Conditional

$$f_{X_1|X_2} = \frac{1/4}{1/2} = \frac{1}{2}$$

5.2.6 (f) Independence

$$\frac{1}{8} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}$$

5.3 Example 3: Trivariate Distribution

$$f_{X,Y,Z} = \begin{cases} Ke^{-(ax+by+cz)}, & x, y, z > 0 \\ 0, & \text{otherwise} \end{cases}$$

5.3.1 (a) Constant

$$K\frac{1}{a}\frac{1}{b}\frac{1}{c} = 1 \Rightarrow K = abc$$

5.3.2 (b) Marginal

$$f_{X,Y} = abe^{-(ax+by)}$$

5.3.3 (c) Marginal

$$f_X = ae^{-ax}$$

5.3.4 (d) Independence

$$abce^{-(ax+by+cz)} = (ae^{-ax})(be^{-by})(ce^{-cz})$$

6 Application: Large MIMO System

6.1 System Setup

M: Transmit antennas
N: Receive antennas

6.2 Channel Matrix

$$H = \begin{bmatrix} h_{11} & \cdots & h_{1N} \\ \vdots & \ddots & \vdots \\ h_{M1} & \cdots & h_{MN} \end{bmatrix}$$